

1. Trigonometric and hyperbolic functions — compare first

Definitions and reciprocal structure

Circular trigonometric

$$\begin{aligned}\tan x &= \frac{\sin x}{\cos x} & \cot x &= \frac{\cos x}{\sin x} \\ \sec x &= \frac{1}{\cos x} & \csc x &= \frac{1}{\sin x}\end{aligned}$$

Hyperbolic

$$\begin{aligned}\sinh x &= \frac{e^x - e^{-x}}{2} & \cosh x &= \frac{e^x + e^{-x}}{2} \\ \tanh x &= \frac{\sinh x}{\cosh x} & \coth x &= \frac{\cosh x}{\sinh x} \\ \operatorname{sech} x &= \frac{1}{\cosh x} & \operatorname{csch} x &= \frac{1}{\sinh x}\end{aligned}$$

Pythagorean-type identities

Circular trigonometric

$$\begin{aligned}\cos^2 x + \sin^2 x &= 1 \\ 1 + \tan^2 x &= \sec^2 x \\ \cot^2 x + 1 &= \csc^2 x\end{aligned}$$

Hyperbolic

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= 1 \\ 1 - \tanh^2 x &= \operatorname{sech}^2 x \\ \coth^2 x - 1 &= \operatorname{csch}^2 x\end{aligned}$$

Parity and addition formulas

Circular trigonometric

$$\begin{aligned}\sin(-x) &= -\sin x & \cos(-x) &= \cos x \\ \tan(-x) &= -\tan x \\ \sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}\end{aligned}$$

Hyperbolic

$$\begin{aligned}\sinh(-x) &= -\sinh x & \cosh(-x) &= \cosh x \\ \tanh(-x) &= -\tanh x \\ \sinh(\alpha \pm \beta) &= \sinh \alpha \cosh \beta \pm \cosh \alpha \sinh \beta \\ \cosh(\alpha \pm \beta) &= \cosh \alpha \cosh \beta \pm \sinh \alpha \sinh \beta \\ \tanh(\alpha \pm \beta) &= \frac{\tanh \alpha \pm \tanh \beta}{1 \pm \tanh \alpha \tanh \beta}\end{aligned}$$

Double-angle and half-angle patterns

Circular trigonometric

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x \\ \cos 2x &= \cos^2 x - \sin^2 x \\ &= 2 \cos^2 x - 1 = 1 - 2 \sin^2 x \\ \tan 2x &= \frac{2 \tan x}{1 - \tan^2 x} \\ \sin^2 \frac{x}{2} &= \frac{1 - \cos x}{2} & \cos^2 \frac{x}{2} &= \frac{1 + \cos x}{2} \\ \tan \frac{x}{2} &= \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}\end{aligned}$$

Hyperbolic

$$\begin{aligned}\sinh 2x &= 2 \sinh x \cosh x \\ \cosh 2x &= \cosh^2 x + \sinh^2 x \\ &= 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x \\ \tanh 2x &= \frac{2 \tanh x}{1 + \tanh^2 x} \\ \sinh^2 \frac{x}{2} &= \frac{\cosh x - 1}{2} & \cosh^2 \frac{x}{2} &= \frac{\cosh x + 1}{2} \\ \tanh \frac{x}{2} &= \frac{\sinh x}{\cosh x + 1} = \frac{\cosh x - 1}{\sinh x}\end{aligned}$$

2. Derivatives and series — matched row by row

Direct derivatives

Circular trigonometric

$$\begin{aligned}\frac{d}{dx}(\sin x) &= \cos x & \frac{d}{dx}(\cos x) &= -\sin x \\ \frac{d}{dx}(\tan x) &= \sec^2 x & \frac{d}{dx}(\cot x) &= -\csc^2 x \\ \frac{d}{dx}(\sec x) &= +\sec x \tan x & \frac{d}{dx}(\csc x) &= -\csc x \cot x\end{aligned}$$

Hyperbolic

$$\begin{aligned}\frac{d}{dx}(\sinh x) &= \cosh x & \frac{d}{dx}(\cosh x) &= +\sinh x \\ \frac{d}{dx}(\tanh x) &= \operatorname{sech}^2 x & \frac{d}{dx}(\coth x) &= -\operatorname{csch}^2 x \\ \frac{d}{dx}(\operatorname{sech} x) &= -\operatorname{sech} x \tanh x & \frac{d}{dx}(\operatorname{csch} x) &= -\operatorname{csch} x \coth x\end{aligned}$$

Inverse-function derivatives

Circular trigonometric

$$\begin{aligned}\frac{d}{dx}(\arcsin x) &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\arccos x) &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\arctan x) &= \frac{1}{1+x^2} \\ \frac{d}{dx}(\operatorname{arccot} x) &= -\frac{1}{1+x^2} \\ \frac{d}{dx}(\operatorname{arcsec} x) &= +\frac{1}{|x|\sqrt{x^2-1}} \\ \frac{d}{dx}(\operatorname{arccsc} x) &= -\frac{1}{|x|\sqrt{x^2-1}}\end{aligned}$$

$|x| < 1$ for arcsin, arccos; $|x| > 1$ for arcsec, arccsc.

Hyperbolic

$$\begin{aligned}\frac{d}{dx}(\operatorname{arsinh} x) &= \frac{1}{\sqrt{1+x^2}} \\ \frac{d}{dx}(\operatorname{arcosh} x) &= +\frac{1}{\sqrt{x^2-1}} \\ \frac{d}{dx}(\operatorname{artanh} x) &= \frac{1}{1-x^2} \\ \frac{d}{dx}(\operatorname{arcoth} x) &= +\frac{1}{1-x^2} \\ \frac{d}{dx}(\operatorname{arcsech} x) &= -\frac{1}{x\sqrt{1-x^2}} \\ \frac{d}{dx}(\operatorname{arcsch} x) &= -\frac{1}{|x|\sqrt{1+x^2}}\end{aligned}$$

$x > 1$ for arcosh; $|x| < 1$ for artanh; $|x| > 1$ for arcoth;
 $0 < x < 1$ for arcsech; $x \neq 0$ for arcsch.

Maclaurin sign patterns

Circular trigonometric

$$\begin{aligned}\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\end{aligned}$$

Hyperbolic

$$\begin{aligned}\sinh x &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \\ \cosh x &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\end{aligned}$$

3. Antiderivatives — the same families compared

Basic and logarithmic antiderivatives

Circular trigonometric

$$\begin{aligned}\int \sin x \, dx &= -\cos x + C \\ \int \cos x \, dx &= \sin x + C \\ \int \sec^2 x \, dx &= \tan x + C \\ \int \csc^2 x \, dx &= -\cot x + C \\ \int \sec x \tan x \, dx &= +\sec x + C \\ \int \csc x \cot x \, dx &= -\csc x + C \\ \int \tan x \, dx &= -\ln |\cos x| + C \\ \int \cot x \, dx &= \ln |\sin x| + C \\ \int \sec x \, dx &= \ln |\sec x + \tan x| + C \\ \int \csc x \, dx &= -\ln |\csc x + \cot x| + C\end{aligned}$$

Hyperbolic

$$\begin{aligned}\int \sinh x \, dx &= \cosh x + C \\ \int \cosh x \, dx &= \sinh x + C \\ \int \operatorname{sech}^2 x \, dx &= \tanh x + C \\ \int \operatorname{csch}^2 x \, dx &= -\coth x + C \\ \int \operatorname{sech} x \tanh x \, dx &= -\operatorname{sech} x + C \\ \int \operatorname{csch} x \coth x \, dx &= -\operatorname{csch} x + C \\ \int \tanh x \, dx &= \ln(\cosh x) + C \\ \int \coth x \, dx &= \ln |\sinh x| + C \\ \int \operatorname{sech} x \, dx &= \arctan(\sinh x) + C \\ \int \operatorname{csch} x \, dx &= \ln \left| \tanh \frac{x}{2} \right| + C\end{aligned}$$

4. Family-specific formulas

Trigonometric: complementary angles and periods

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\sin(x + 2\pi) = \sin x \quad \cos(x + 2\pi) = \cos x$$

$$\tan(x + \pi) = \tan x$$

Trigonometric: triple-angle and half-angle substitution

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\sin x = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)}$$

$$\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$$

Trigonometric: product-to-sum

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

$$\sin \alpha \sin \beta = -\frac{1}{2} [\cos(\alpha + \beta) - \cos(\alpha - \beta)]$$

Special angles

θ	18°	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	$\frac{\sqrt{5}-1}{4}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	$\frac{\sqrt{10+2\sqrt{5}}}{4}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	$\frac{1}{\sqrt{5+2\sqrt{5}}}$	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef
$\cot \theta$	$\sqrt{5+2\sqrt{5}}$	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0
$\sec \theta$	$\frac{4}{\sqrt{10+2\sqrt{5}}}$	$\frac{2\sqrt{3}}{3}$	$\sqrt{2}$	2	undef
$\csc \theta$	$\sqrt{5+1}$	2	$\sqrt{2}$	$\frac{2\sqrt{3}}{3}$	1

Trigonometric: sum-to-product

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

Inverse hyperbolic logarithmic forms

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{1+x^2}\right)$$

$$\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2-1}\right) \quad x \geq 1$$

$$\operatorname{artanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad |x| < 1$$

$$\operatorname{arcsech} x = \ln\left(\frac{1 + \sqrt{1-x^2}}{x}\right) \quad 0 < x \leq 1$$

$$\operatorname{arccsch} x = \ln\left(\frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right) \quad x \neq 0$$

5. General differentiation, integration, and Taylor formulas

Differentiation: core rules

$$\begin{aligned}\frac{d}{dx}(c) &= 0 & \frac{d}{dx}(x) &= 1 \\ \frac{d}{dx}[cf(x)] &= cf'(x) \\ \frac{d}{dx}[f(x) \pm g(x)] &= f'(x) \pm g'(x) \\ \frac{d}{dx}[f(x)g(x)] &= f'(x)g(x) + f(x)g'(x) \\ \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] &= \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)} \\ \frac{d}{dx}f(g(x)) &= f'(g(x))g'(x)\end{aligned}$$

Derivative of an inverse function

$$f(f^{-1}(x)) = x \implies f'(f^{-1}(x))(f^{-1})'(x) = 1$$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$\frac{dy}{dx} = \frac{1}{dx/dy} \quad y = f^{-1}(x)$$

Powers, exponentials, logarithms

$$\begin{aligned}\frac{d}{dx}(x^n) &= nx^{n-1} & \frac{d}{dx}\left(\frac{1}{x}\right) &= -\frac{1}{x^2} \\ \frac{d}{dx}\sqrt{x} &= \frac{1}{2\sqrt{x}} & \frac{d}{dx}(e^x) &= e^x \\ \frac{d}{dx}(a^x) &= a^x \ln a & \frac{d}{dx} \ln |x| &= \frac{1}{x} \\ \frac{d}{dx} \log_a |x| &= \frac{1}{x \ln a}\end{aligned}$$

Taylor's theorem

$$f(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1},$$

where ξ lies between x and a .

Integration: core rules

$$\begin{aligned}\int [cf(x) \pm g(x)] dx &= c \int f(x) dx \pm \int g(x) dx \\ \int f(x)g'(x) dx &= f(x)g(x) - \int f'(x)g(x) dx\end{aligned}$$

The second line is integration by parts.

Powers, exponentials, logarithms

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad n \neq -1 \\ \int \frac{1}{x} dx &= \ln |x| + C \\ \int e^x dx &= e^x + C \\ \int a^x dx &= \frac{a^x}{\ln a} + C \\ \int \ln |x| dx &= x \ln |x| - x + C\end{aligned}$$

Standard inverse-trig and logarithmic forms

$$\begin{aligned}\int \frac{1}{x^2 + a^2} dx &= \frac{1}{a} \arctan \frac{x}{a} + C \\ \int \frac{1}{a^2 - x^2} dx &= \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C \\ \int \frac{1}{x^2 - a^2} dx &= \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \\ \int \frac{1}{\sqrt{a^2 - x^2}} dx &= \arcsin \frac{x}{a} + C \\ \int \frac{1}{\sqrt{x^2 + a^2}} dx &= \ln \left| x + \sqrt{x^2 + a^2} \right| + C \\ \int \frac{1}{\sqrt{x^2 - a^2}} dx &= \ln \left| x + \sqrt{x^2 - a^2} \right| + C \\ &a > 0\end{aligned}$$

Other common Maclaurin series

$$\begin{aligned}e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\ \frac{1}{1-x} &= 1 + x + x^2 + x^3 + \dots \\ \frac{1}{1+x} &= 1 - x + x^2 - x^3 + \dots \\ \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \\ \arctan x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \\ \operatorname{artanh} x &= x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \\ (1+x)^\alpha &= 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 \\ &\quad + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + \dots\end{aligned}$$

$|x| < 1$ for the geometric, logarithmic, inverse-tangent, inverse-hyperbolic-tangent, and binomial series.